

Introduction

Data analysis and vizualization in Python - RSE course

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RSE = Research software engineering

- **Software development & maintenance** for research projects
- **Training and workshops** on programming and research software skills
- **Consultations** for research projects and grant proposals
- **Support for HPC computing** (e.g. Chimera cluster)
- **Development of research tools and platforms** (e.g. AI Sandbox)
- contact: rse.cuni.cz
- MY ROLE:
 - focus on LLMs, AI in general and AI agents
 - finished master studies at CTU - Software engineering (Artificial intelligence)
- YOUR ROLE?

Course overview

- *1-day hands-on practical Data analysis and vizualization in Python for beginners & researchers*
- **Course time plan**
- **sticky notes**

13:10 – 13:30	Introduction & setup
13:30 – 14:00	loading datasets & dataframes
14:00 – 14:40	dataframes filtering & aggregation
14:40 – 15:00	break
15:00 – 15:30	errors in data
15:30 – 16:30	Matplotlib
16:30 – 17:00	project

Course materials

- Jupyter notebooks and slides on github

<https://github.com/rse-cuni/training-materials/tree/main/Data-analysis-Python>

- Jupyter notebooks available on chimera
- connect to chimera:
 - <https://hpc.troja.mff.cuni.cz:8000>
 - You will be asked for credentials.
 - Put in your SIS login name and password
 - You should end up here:
 - Press Start

Server Options

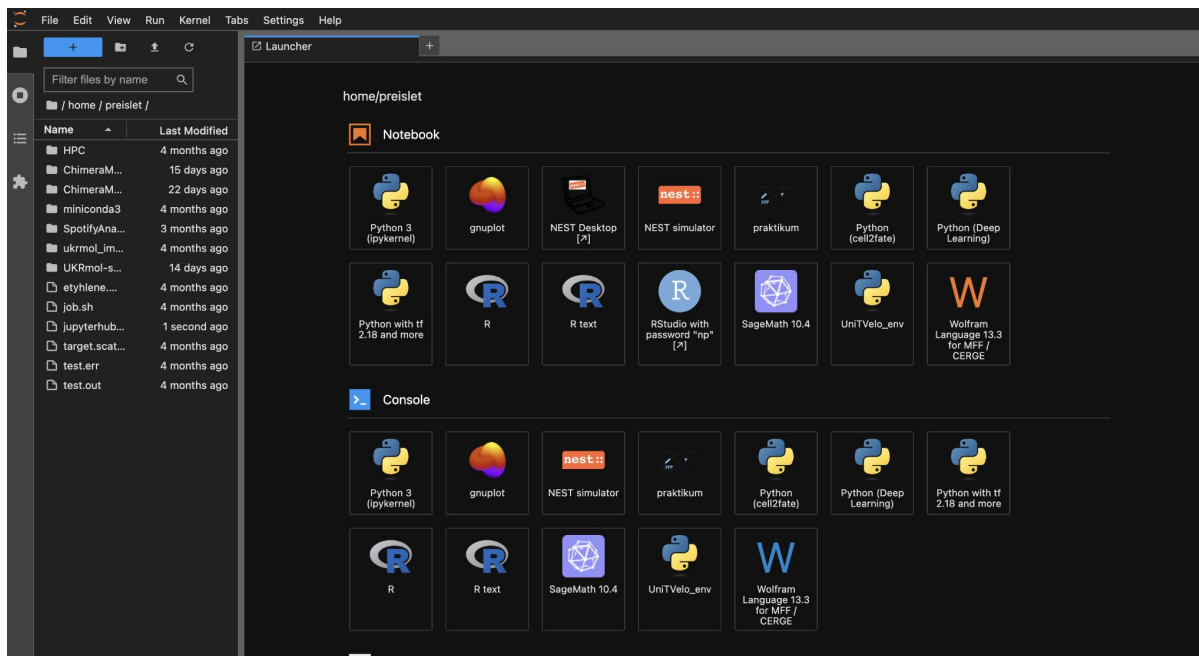
Large part of the cluster will be moved on 2026-04-09.
The cluster will be down on that day (and possibly for few days afterwards). We will restore service as soon as possible.
We apologize for the inconvenience.

Partition:	edu	Partition	Time limit	Usage
Time:	2:00:00	edu	3h	education / classes less demanding jobs
Memory:	2G	gpu-ffa	12h	jobs that require GPU (L40)
CPUs per task:	2	ffa	12h	low priority, HPC jobs
Account:		<i>For more partitions (for departments, preemptable,...) and more details please visit our gitlab.</i>		
Other SLURM options:				

Start

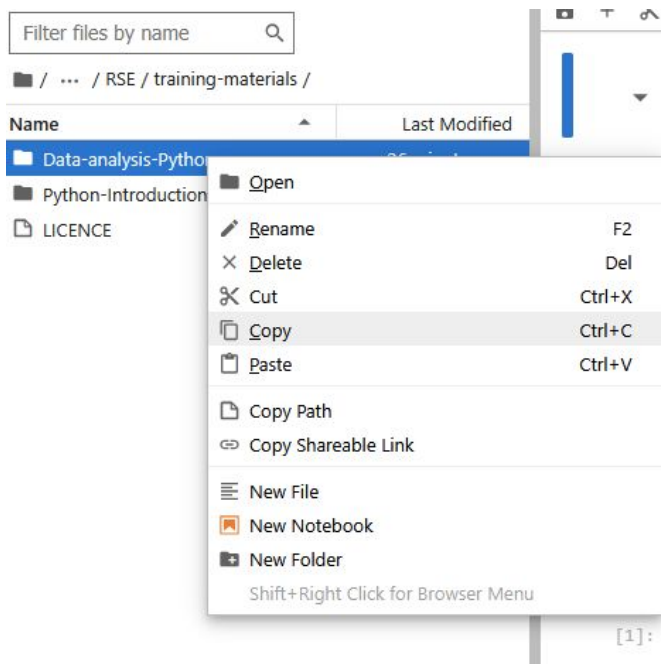
Chimera JupyterLab setup

- go to /home/teaching/RSE/training-materials



Chimera JupyterLab setup

- right-click on Data-analysis-Python and click Copy



Chimera JupyterLab setup

- go back to your home directory:
/home/your_cas_login
- and right click and Paste

